

T5.1

Adversarial Perturbation

SAM

S1 scope: base optimizer wrapper
S2 signal: gradient at perturbed weights
S3 state: inherited base-optimizer state
S4 reconstruct: identity reconstruction
S5 update: outer step using perturbed gradient

LookSAM

+ Reuse ascent
S2: reused ascent direction

GCSAM

+ Filter signal
S2: filtered or centralized perturbation

bSAM

+ Reweight sharpness
S2/S5: weighted or Bayesian sharpness

ASAM

+ Rescale neighborhood
S2: scale-invariant perturbation

GSAM

+ Decompose direction
S2/S5: surrogate-gap direction

T5.2

Diagonal Hessian Estimation

Sophia

S1 scope: element-wise parameters
S2 signal: gradient plus Hessian/Gauss-Newton probe
S3 state: diagonal curvature proxy h_t
S4 reconstruct: identity reconstruction
S5 update: clipped curvature-preconditioned ratio

AdaHessian

+ Clip ratio
S3: Hutchinson diagonal Hessian

Sophia-H/-G

S3: HVP/gradient-based proxy

T5.3

Post-Processing & Selective Filtering

Cautious Optimizers

S1 scope: inherited base optimizer
S2 signal: current gradient for agreement
S3 state: base momentum state
S4 reconstruct: candidate update unchanged
S5 update: mask coordinates opposing gradient

SPAM / StableSPAM

S3/S5: spike reset + norm stabilization

StableAdamW

S5: update clipping / stable denominator

Magma / MGUP

+ Relax mask
S5: selective aligned updates

GC

S2: zero-mean gradient projection

AdamP

S5: remove radial component

Grams

S5: gradient direction + momentum scale

T5.4

Layer-Wise Trust Region Scaling

LAMB

S1 scope: layer or parameter groups
S2 signal: Adam-style gradient signal
S3 state: Adam first/second moments
S4 reconstruct: full group update
S5 update: layer-wise trust-ratio scaling

AGC

S1/S5: parameter-relative gradient clipping

LARS

Trust-ratio predecessor